

9 (long)

(a) X and Y are two 2-bit unsigned binary numbers, where $Y > 1$. The 4-bit binary number P satisfies the equation:

$$P = (X)^2 + Y$$

Design a digital circuit that calculates the most significant bit and the least significant bit of P from the bits of X and Y . The circuit should be composed only of NOR gates and the smallest possible number of such gates should be used. [15]

(b) 2-bit binary unsigned numbers are received sequentially by a detector. Each of these numbers is greater than 1. The output Z of the detector becomes 1 if the current value of the number received is greater than 2 and the value of the previous number received was equal to 2. The output Z is otherwise 0.

(i) Draw a state diagram for the system. [6]

(ii) State how many JK bistables will be needed to implement the detector and draw a state transition table (assume that unused states are never reached). [5]

(iii) Derive simplified Boolean expressions for the J inputs of the bistables. [4]